Daily Problem November 28, 2025

Find:

$$\int_0^{\pi/2} \ln(\cos x)dx$$

Solution:

$$I = \int_0^{\pi/2} \ln(\cos x)dx.$$

let $x = \frac{\pi}{2} - t$,we can get:

$$I = \int_{\pi/2}^{0} \ln\left(\cos\left(\frac{\pi}{2} - t\right)\right)(-dt) = \int_0^{\pi/2} \ln(\sin t)dt = \int_0^{\pi/2} \ln(\sin x)dx.$$

$$2I = \int_0^{\pi/2} \ln(\cos x)dx + \int_0^{\pi/2} \ln(\sin x)dx = \int_0^{\pi/2} \ln(\cos x \sin x)dx = \int_0^{\pi/2} \ln(\frac{\sin 2x}{2})$$

thus,we can get:

$$2I = \int_0^{\pi/2} \ln(\sin 2x)dx - \int_0^{\pi/2} \ln 2dx.$$

use $u = 2x$, we can get:

$$\int_0^{\pi/2} \ln(\sin 2x)dx = \int_0^{\pi} \ln(\sin u) \cdot \frac{du}{2} = \frac{1}{2}\int_0^{\pi} \ln(\sin u)du.$$

$$\int_0^{\pi} \ln(\sin u)du = \int_0^{\pi/2} \ln(\sin u)du + \int_{\pi/2}^{\pi} \ln(\sin u)du = 2I,$$

Therefore,

$$\int_0^{\pi/2} \ln(\sin 2x)dx = \frac{1}{2} \cdot 2I = I.$$

and

$$2I = I - \frac{\pi}{2}\ln 2.$$

Hence, we conclude that

$$\int_0^{\pi/2} \ln(\cos x)dx = -\frac{\pi}{2}\ln 2.$$